Chapter 1 Duolingo Math checklist

Tick each textbook section as you read it and each Duolingo unit as you finish it. Units are listed per lane (a Duolingo grade or topic) in the order the app presents them, so you can follow one lane top to bottom. Ratings: full = covers the section's main skill, partial = covers part of it, prereq = a warm-up from an earlier grade. The right column names the textbook sections the unit serves.

Chapter 1: Functions and Their Graphs (p. 1)

Grade 9 carries the basics (functions, lines, transformations), grade 11 repeats them at precalculus depth, and the Algebraic Graphing topic is a condensed walk through the whole chapter. Follow grade 11 if you only have time for one lane.

Textbook sections

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☐	1.10 Mathematical Modeling and Variation	p. 102

Book only, not on Duolingo: **1.1** Translating points in the plane; Applications of the distance and midpoint formulas. **1.2** Symmetry tests (x-axis, y-axis, origin); Sketching graphs of general equations by point plotting. **1.3** Slope as a rate of change in applications. **1.4** The difference quotient; Writing a function from a word problem. **1.5** Even and odd functions; The Vertical Line Test. **1.6** The square root and reciprocal parent functions in detail. **1.8** Domain of a composite function; Decomposing a function into two simpler functions. **1.9** One-to-one functions and the Horizontal Line Test. **1.10** Inverse, joint, and combined variation; Least squares regression on the calculator.

Grade 6

☐	33. Plot points in all quadrants	prereq	1.1

Grade 7

☐	7. Proportionality in equations	prereq	1.10

Grade 8

☐	21. Intro to slope	prereq	1.3
☐	22. Slope formula	prereq	1.3
☐	24. Slope-intercept form	prereq	1.3
☐	28. Equations of parallel lines	full	1.3
☐	29. Equations of perpendicular lines	full	1.3
☐	36. Intro to functions	prereq	1.4

Grade 9

☐	5. Slope-intercept form of linear equations	full	1.3
☐	7. Standard form of linear equations	full	1.3
☐	17. Inputs and outputs	full	1.4
☐	18. Graphs of functions	full	1.5
☐	20. Vertical intercept of functions	partial	1.2
☐	21. Horizontal intercept of functions	partial	1.2
☐	23. Average rate of change	full	1.5
☐	24. Linear functions	prereq	1.6
☐	25. Domain of a function	full	1.4
☐	26. Range of a function	partial	1.4
☐	27. Graphs of piecewise functions	full	1.5
☐	28. Equations of piecewise functions	full	1.4
☐	30. Step functions	full	1.6
☐	31. Absolute value functions	full	1.6
☐	33. Graphs of inverse functions	partial	1.9
☐	34. Equations of inverse functions	partial	1.9
☐	35. Graphs of inverse linear functions	prereq	1.9
☐	36. Equations of inverse linear functions	prereq	1.9
☐	75. Vertical translations	full	1.7
☐	76. Horizontal translations	full	1.7
☐	77. Vertical stretches	full	1.7
☐	78. Vertical compressions	full	1.7
☐	79. Horizontal stretches	full	1.7
☐	80. Horizontal compressions	full	1.7
☐	81. Scatter plots	partial	1.10
☐	82. Correlation	partial	1.10
☐	83. Linear models	partial	1.10
☐	84. Residuals	partial	1.10

Grade 10

☐	10. Reflectional symmetry	prereq	1.2
☐	45. Lines in point-slope form	full	1.3
☐	46. Lines in different forms	full	1.3
☐	47. Circles on the plane	full	1.2
☐	50. Line segment midpoint	full	1.1
☐	51. Distance formula	full	1.1

Grade 11

☐	2. Function families	full	1.6
☐	3. Compare function graphs	full	1.5
☐	9. Transform absolute value functions	full	1.7
☐	13. Relative maxima and minima of polynomial functions	partial	1.5
☐	65. Inverse function equations	full	1.9
☐	66. Inverse function graphs	full	1.9
☐	67. Combine functions	full	1.8
☐	68. Function composition	full	1.8
☐	69. Vertical translations	full	1.7
☐	70. Horizontal translations	full	1.7
☐	71. X-axis function reflections	full	1.7
☐	72. Y-axis function reflections	full	1.7
☐	73. Vertical stretches	full	1.7
☐	74. Vertical compressions	full	1.7
☐	75. Horizontal stretches	full	1.7
☐	76. Horizontal compressions	full	1.7

Grade 12

☐	13. Function composition	full	1.8
☐	14. Inverse functions	full	1.9
☐	17. Domain and range	full	1.5
☐	18. Graphing inverses	full	1.9

Geometry

☐	11. Absolute value and distance	prereq	1.1

Algebraic Graphing

☐	3. Slope	partial	1.3
☐	4. Slope-intercept lines	partial	1.3
☐	8. Function notation	full	1.4
☐	9. Piecewise-defined functions	full	1.4
☐	10. Step and absolute value functions	full	1.6
☐	23. Inverse linear functions	partial	1.9
☐	24. Function translations	full	1.7
☐	25. Vertical function scaling	full	1.7
☐	26. Horizontal function scaling	full	1.7
☐	27. Reflections of functions	full	1.7

Statistics

☐	15. Linear models	partial	1.10
☐	16. Linear model fitness	partial	1.10

Section guide

For each section: the Duolingo units that practise it (already listed above with checkboxes), then the parts only the textbook teaches. Tick each book-only item once you have worked it in the book.

1.1 Rectangular Coordinates (p. 2)

Distance and midpoint are covered in grade 10. Plotting is a lower-grade prerequisite. The section's data applications are book only.

Duolingo, in app order:
Grade 6: 33. Plot points in all quadrants (prereq)
Grade 10: 50. Line segment midpoint (full) · 51. Distance formula (full)
Geometry: 11. Absolute value and distance (prereq)

Book only:

- [ ] Translating points in the plane
- [ ] Applications of the distance and midpoint formulas

1.2 Graphs of Equations (p. 13)

Intercepts and circles are covered; symmetry tests and point-plotting of general equations are book only.

Duolingo, in app order:
Grade 9: 20. Vertical intercept of functions (partial) · 21. Horizontal intercept of functions (partial)
Grade 10: 10. Reflectional symmetry (prereq) · 47. Circles on the plane (full)

Book only:

- [ ] Symmetry tests (x-axis, y-axis, origin)
- [ ] Sketching graphs of general equations by point plotting

1.3 Linear Equations in Two Variables (p. 24)

Well covered. Grade 8 builds slope and parallel/perpendicular lines, grade 9 the forms of a line, grade 10 point-slope form.

Duolingo, in app order:
Grade 8: 21. Intro to slope (prereq) · 22. Slope formula (prereq) · 24. Slope-intercept form (prereq) · 28. Equations of parallel lines (full) · 29. Equations of perpendicular lines (full)
Grade 9: 5. Slope-intercept form of linear equations (full) · 7. Standard form of linear equations (full)
Grade 10: 45. Lines in point-slope form (full) · 46. Lines in different forms (full)
Algebraic Graphing: 3. Slope (partial) · 4. Slope-intercept lines (partial)

Book only:

- [ ] Slope as a rate of change in applications

1.4 Functions (p. 39)

Function definition, notation, domain, and piecewise functions are covered. The difference quotient is book only.

Duolingo, in app order:
Grade 8: 36. Intro to functions (prereq)
Grade 9: 17. Inputs and outputs (full) · 25. Domain of a function (full) · 26. Range of a function (partial) · 28. Equations of piecewise functions (full)
Algebraic Graphing: 8. Function notation (full) · 9. Piecewise-defined functions (full)

Book only:

- [ ] The difference quotient
- [ ] Writing a function from a word problem

1.5 Analyzing Graphs of Functions (p. 54)

Domain and range from graphs, zeros, relative extrema, and average rate of change are covered. Even and odd functions and the vertical line test are book only.

Duolingo, in app order:
Grade 9: 18. Graphs of functions (full) · 23. Average rate of change (full) · 27. Graphs of piecewise functions (full)
Grade 11: 3. Compare function graphs (full) · 13. Relative maxima and minima of polynomial functions (partial)
Grade 12: 17. Domain and range (full)

Book only:

- [ ] Even and odd functions
- [ ] The Vertical Line Test

1.6 A Library of Parent Functions (p. 66)

Grade 11's Function families is the closest match to the parent-function library. Step and absolute value functions are drilled separately.

Duolingo, in app order:
Grade 9: 24. Linear functions (prereq) · 30. Step functions (full) · 31. Absolute value functions (full)
Grade 11: 2. Function families (full)
Algebraic Graphing: 10. Step and absolute value functions (full)

Book only:

- [ ] The square root and reciprocal parent functions in detail

1.7 Transformations of Functions (p. 73)

Fully covered. Grade 9 drills each transformation; grade 11 repeats them with reflections added. Do grade 11 if time is short.

Duolingo, in app order:
Grade 9: 75. Vertical translations (full) · 76. Horizontal translations (full) · 77. Vertical stretches (full) · 78. Vertical compressions (full) · 79. Horizontal stretches (full) · 80. Horizontal compressions (full)
Grade 11: 9. Transform absolute value functions (full) · 69. Vertical translations (full) · 70. Horizontal translations (full) · 71. X-axis function reflections (full) · 72. Y-axis function reflections (full) · 73. Vertical stretches (full) · 74. Vertical compressions (full) · 75. Horizontal stretches (full) · 76. Horizontal compressions (full)
Algebraic Graphing: 24. Function translations (full) · 25. Vertical function scaling (full) · 26. Horizontal function scaling (full) · 27. Reflections of functions (full)

1.8 Combinations of Functions: Composite Functions (p. 83)

Arithmetic combinations and composition are covered. Finding the domain of a composite and decomposing a function are lighter in the app than in the book.

Duolingo, in app order:
Grade 11: 67. Combine functions (full) · 68. Function composition (full)
Grade 12: 13. Function composition (full)

Book only:

- [ ] Domain of a composite function
- [ ] Decomposing a function into two simpler functions

1.9 Inverse Functions (p. 92)

Finding and graphing inverses is covered in grades 9, 11, and 12. The horizontal line test and one-to-one functions are book only.

Duolingo, in app order:
Grade 9: 33. Graphs of inverse functions (partial) · 34. Equations of inverse functions (partial) · 35. Graphs of inverse linear functions (prereq) · 36. Equations of inverse linear functions (prereq)
Grade 11: 65. Inverse function equations (full) · 66. Inverse function graphs (full)
Grade 12: 14. Inverse functions (full) · 18. Graphing inverses (full)
Algebraic Graphing: 23. Inverse linear functions (partial)

Book only:

- [ ] One-to-one functions and the Horizontal Line Test

1.10 Mathematical Modeling and Variation (p. 102)

Least squares and linear models are covered in grade 9 and the Statistics topic. Direct variation has a grade 7 prerequisite; inverse and joint variation are book only.

Duolingo, in app order:
Grade 7: 7. Proportionality in equations (prereq)
Grade 9: 81. Scatter plots (partial) · 82. Correlation (partial) · 83. Linear models (partial) · 84. Residuals (partial)
Statistics: 15. Linear models (partial) · 16. Linear model fitness (partial)

Book only:

- [ ] Inverse, joint, and combined variation
- [ ] Least squares regression on the calculator

Textbook: Ron Larson and Robert P. Hostetler. Precalculus with Limits, 2nd ed. (Pre-AP National Student Edition) Thomson Brooks/Cole, 2009. ISBN 978-0-538-73659-6. Duolingo inventory: 820 units, last verified 2026-10-07 on iOS. Only titles and unit names are reproduced, for reference; no textbook or Duolingo content is included. This checklist is licensed CC BY 4.0; see the repository NOTICE for third-party material.